

CHAPTER 1:

```
Welcome to Cloud Shell! Type "help" to get started, or type "gemini" to try prompting with Gemini CLI.
To set your Cloud Platform project in this session use `gcloud config set project [PROJECT_ID]`.
You can view your projects by running `gcloud projects list`.
yalinib884@cloudshell:~$ uname
Linux
yalinib884@cloudshell:~$ uptime
10:26:38 up 4 min, 0 user, load average: 0.86, 0.46, 0.18
yalinib884@cloudshell:~$ hostname
cs-8526177146-default
yalinib884@cloudshell:~$ dmesg
dmesg: read kernel buffer failed: Operation not permitted
yalinib884@cloudshell:~$ free
              total        used        free      shared  buff/cache   available
Mem:           8138088      2238232      4282796          1144       1865184       5899856
Swap:              0              0              0
o yalinib884@cloudshell:~$
```

```
top - 10:35:49 up 13 min, 0 user, load average: 0.00, 0.07, 0.09
Tasks: 34 total, 1 running, 33 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.8 us, 0.8 sy, 0.0 ni, 98.3 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7947.4 total, 4186.2 free, 2179.5 used, 1824.0 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 5767.9 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1545	yalinib+	20	0	1145620	76748	48520	S	0.7	0.9	0:02.50	node
449	root	20	0	1234268	12312	8288	S	0.3	0.2	0:00.54	editor-proxy
1	root	20	0	4324	3432	3152	S	0.0	0.0	0:00.07	bash
8	syslog	20	0	222140	3784	2440	S	0.0	0.0	0:00.01	rsyslogd
24	root	20	0	38768	28004	10896	S	0.0	0.3	0:00.64	python
25	root	20	0	6196	2696	2456	S	0.0	0.0	0:00.00	logger
66	root	10	-10	12024	4224	3108	S	0.0	0.1	0:00.00	sshd
384	root	20	0	2022092	85700	62440	S	0.0	1.1	0:00.44	dockerd
450	root	20	0	16328	6416	5488	S	0.0	0.1	0:00.01	sudo
454	root	20	0	1226100	2680	1976	S	0.0	0.0	0:00.00	tmux-agent
491	root	20	0	1793224	46272	32412	S	0.0	0.6	0:00.98	containerd
497	root	10	-10	13996	9744	8364	S	0.0	0.1	0:00.02	sshd
514	root	10	-10	13996	9736	8356	S	0.0	0.1	0:00.03	sshd
540	yalinib+	10	-10	14256	6368	4708	S	0.0	0.1	0:00.00	sshd
563	yalinib+	10	-10	7340	3724	3476	S	0.0	0.0	0:00.00	bash
566	yalinib+	10	-10	14256	6424	4764	S	0.0	0.1	0:00.00	sshd
569	yalinib+	10	-10	7604	4440	3836	S	0.0	0.1	0:00.00	bash

```
To set your Cloud Platform project in this session use `gcloud config set project [PROJECT_ID]`.
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yalinib884@cloudshell:~$ lsb_release
No LSB modules are available.
yalinib884@cloudshell:~$ arch
x86_64
yalinib884@cloudshell:~$ uname
Linux
```

CHAPTER 2:

```
yalinib884@cloudshell:~$ mkdir new_folder
yalinib884@cloudshell:~$ cd new_folder
yalinib884@cloudshell:~/new_folder$ touch file.txt
yalinib884@cloudshell:~/new_folder$ ls
file.txt
yalinib884@cloudshell:~/new_folder$ touch file1.txt file3.txt
yalinib884@cloudshell:~/new_folder$ ls
file1.txt  file3.txt  file.txt
yalinib884@cloudshell:~/new_folder$ cp file1.txt copy_file1.txt
yalinib884@cloudshell:~/new_folder$ ls
copy_file1.txt  file1.txt  file3.txt  file.txt
yalinib884@cloudshell:~/new_folder$ mv file2.txt newfile.txt
mv: cannot stat 'file2.txt': No such file or directory
yalinib884@cloudshell:~/new_folder$ mv file3.txt newfile.txt
yalinib884@cloudshell:~/new_folder$ ls
copy_file1.txt  file1.txt  file.txt  newfile.txt
yalinib884@cloudshell:~/new_folder$ ln file1.txt hardlink.txt
bash: syntax error near unexpected token `ln'
yalinib884@cloudshell:~/new_folder$ ln file1.txt harderlink.txt
yalinib884@cloudshell:~/new_folder$ ls -l
-rw-r--r-- 2 yalinib884 yalinib884 0 Oct 10 14:54 copy_file1.txt
-rw-r--r-- 2 yalinib884 yalinib884 0 Oct 10 14:54 file1.txt
-rw-r--r-- 1 yalinib884 yalinib884 0 Oct 10 14:54 file.txt
-rw-r--r-- 1 yalinib884 yalinib884 0 Oct 10 14:54 harderlink.txt
-rw-r--r-- 1 yalinib884 yalinib884 0 Oct 10 14:54 newfile.txt
yalinib884@cloudshell:~/new_folder$ rm file1.txt
yalinib884@cloudshell:~/new_folder$ cd..
```

CHAPTER 3:

```
yalinib884@cloudshell:~$ mkdir linux_new
yalinib884@cloudshell:~$ cd linux_new
yalinib884@cloudshell:~/linux_new$ touch notes.txt
yalinib884@cloudshell:~/linux_new$ nano notes.txt
yalinib884@cloudshell:~/linux_new$ cat notes.txt
Hello Linux
this is a line1
this is a line2
this is a line3

yalinib884@cloudshell:~/linux_new$ ls
notes.txt
yalinib884@cloudshell:~/linux_new$ more notes.txt
Hello Linux
this is a line1
this is a line2
this is a line3

yalinib884@cloudshell:~/linux_new$
yalinib884@cloudshell:~/linux_new$ less note.txt
note.txt: No such file or directory
yalinib884@cloudshell:~/linux_new$ less notes.txt
yalinib884@cloudshell:~/linux_new$ head notes.txt
Hello Linux
this is a line1
this is a line2
this is a line3
```

```
yalinib884@cloudshell:~/linux_new$ head notes.txt
Hello Linux
this is a line1
this is a line2
```

```
yalinib884@cloudshell:~/linux_new$ vim notes.txt
yalinib884@cloudshell:~/linux_new$ cat notes.txt
This Linux os
Hello Linux
this is a line1
this is a line2
this is a line3
This is end of the line

yalinib884@cloudshell:~/linux_new$ get notes.txt
-bash: get: command not found
yalinib884@cloudshell:~/linux_new$ gedit notes.txt
-bash: gedit: command not found
yalinib884@cloudshell:~/linux_new$ awk '{print $1}' notes.txt
This
Hello
this
this
this
This
```

```

yalinib884@cloudshell:~/linux_new$ awk '{print NR,$0}' notes.txt
1 This Linux os
2 Hello Linux
3 this is a line1
4 this is a line2
5 this is a line3
6 This is end of the line
7
8
yalinib884@cloudshell:~/linux_new$ sed 's/Linux/Uduntu/' notes.txt
This Uduntu os
Hello Uduntu
this is a line1
this is a line2
this is a line3
This is end of the line

yalinib884@cloudshell:~/linux_new$ sed -i 's/Linux/Ubuntu/g' notes.txt
yalinib884@cloudshell:~/linux_new$ cat notes.txt
This Ubuntu os
Hello Ubuntu
this is a line1
this is a line2
this is a line3
This is end of the line

```

CHAPTER 4:

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yalinib884@cloudshell:~/linux_new$ mkdir permission_practie
yalinib884@cloudshell:~/linux_new$ cd permission_practie
yalinib884@cloudshell:~/linux_new/permission_practie$ touch demo.txt
yalinib884@cloudshell:~/linux_new/permission_practie$ ls -l
demo.txt
yalinib884@cloudshell:~/linux_new/permission_practie$ change +x demo.txt
-bash: change: command not found
yalinib884@cloudshell:~/linux_new/permission_practie$ chmod +x demo.txt
yalinib884@cloudshell:~/linux_new/permission_practie$ ls -l
demo.txt
yalinib884@cloudshell:~/linux_new/permission_practie$ chmod 777 demo.txt
yalinib884@cloudshell:~/linux_new/permission_practie$ ls -l
demo.txt
yalinib884@cloudshell:~/linux_new/permission_practie$ sudo chown root demo.txt
yalinib884@cloudshell:~/linux_new/permission_practie$ ls -l
demo.txt
yalinib884@cloudshell:~/linux_new/permission_practie$ sudo chgrp root demo.txt
yalinib884@cloudshell:~/linux_new/permission_practie$ umask
0002
yalinib884@cloudshell:~/linux_new/permission_practie$ touch test.txt
yalinib884@cloudshell:~/linux_new/permission_practie$ ls -l
demo.txt
test.txt

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test.txt
yalinib884@cloudshell:~/linux_new/permission_practie$ stat demo.txt
  File: demo.txt
  Size: 0          Blocks: 0          IO Block: 4096   regular empty file
Device: 8,17   Inode: 131505   Links: 1
Access: (0777/-rwxrwxrwx)  Uid: (   0/   root)   Gid: (   0/   root)
Access: 2026-05-14 17:37:05.706331600 +0000
Modify: 2026-05-14 17:37:05.706331600 +0000
Change: 2026-05-14 17:39:15.899158926 +0000
 Birth: 2026-05-14 17:37:05.706331600 +0000
yalinib884@cloudshell:~/linux_new/permission_practie$ sudo usermod -aG sudo username
usermod: user 'username' does not exist
yalinib884@cloudshell:~/linux_new/permission_practie$

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CHAPTER 5:

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usermod: user 'username' does not exist
yalinib884@cloudshell:~/linux_new/permission_practie$ ifconfig
docker0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1460
    inet 172.17.0.1 netmask 255.255.0.0 broadcast 172.17.255.255
    ether b6:86:e2:37:7b:87 txqueuelen 0 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1460
    inet 10.88.0.3 netmask 255.255.0.0 broadcast 10.88.255.255
    ether 52:52:f9:e7:06:7e txqueuelen 1000 (Ethernet)
    RX packets 51700 bytes 19884595 (19.8 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 40020 bytes 58683902 (58.6 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 59913 bytes 123905507 (123.9 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 59913 bytes 123905507 (123.9 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

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yalinib884@cloudshell:~/linux_new/permission_practie$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
2: eth0@if7: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1460 qdisc noqueue state UP group default qlen 1000
    link/ether 52:52:f9:e7:06:7e brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 10.88.0.3/16 brd 10.88.255.255 scope global eth0
        valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1460 qdisc noqueue state DOWN group default
    link/ether b6:86:e2:37:7b:87 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever
yalinib884@cloudshell:~/linux_new/permission_practie$ ping google.com

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        valid_lft forever preferred_lft forever
yalinib884@cloudshell:~/linux_new/permission_practie$ ping google.com
PING google.com (142.251.10.100) 56(84) bytes of data.
64 bytes from sd-in-f100.1e100.net (142.251.10.100): icmp_seq=1 ttl=114 time=0.789 ms
64 bytes from sd-in-f100.1e100.net (142.251.10.100): icmp_seq=2 ttl=114 time=0.576 ms
64 bytes from sd-in-f100.1e100.net (142.251.10.100): icmp_seq=3 ttl=114 time=0.576 ms
64 bytes from sd-in-f100.1e100.net (142.251.10.100): icmp_seq=4 ttl=114 time=0.703 ms
64 bytes from sd-in-f100.1e100.net (142.251.10.100): icmp_seq=5 ttl=114 time=0.517 ms
64 bytes from sd-in-f100.1e100.net (142.251.10.100): icmp_seq=6 ttl=114 time=0.560 ms
64 bytes from sd-in-f100.1e100.net (142.251.10.100): icmp_seq=7 ttl=114 time=0.576 ms
64 bytes from sd-in-f100.1e100.net (142.251.10.100): icmp_seq=8 ttl=114 time=0.792 ms
64 bytes from sd-in-f100.1e100.net (142.251.10.100): icmp_seq=9 ttl=114 time=0.497 ms
64 bytes from sd-in-f100.1e100.net (142.251.10.100): icmp_seq=10 ttl=114 time=0.584 ms
64 bytes from sd-in-f100.1e100.net (142.251.10.100): icmp_seq=11 ttl=114 time=0.573 ms
64 bytes from sd-in-f100.1e100.net (142.251.10.100): icmp_seq=12 ttl=114 time=0.574 ms
64 bytes from sd-in-f100.1e100.net (142.251.10.100): icmp_seq=13 ttl=114 time=0.658 ms
64 bytes from sd-in-f100.1e100.net (142.251.10.100): icmp_seq=14 ttl=114 time=0.624 ms
64 bytes from sd-in-f100.1e100.net (142.251.10.100): icmp_seq=15 ttl=114 time=0.562 ms
64 bytes from sd-in-f100.1e100.net (142.251.10.100): icmp_seq=16 ttl=114 time=0.589 ms

```

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yalinib884@cloudshell:~/linux_new/permission_practie$ netstat -tulnp
(Not all processes could be identified, non-owned process info
 will not be shown, you would have to be root to see it all.)
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 127.0.0.1:46865        0.0.0.0:*                LISTEN      -
tcp        0      0 127.0.0.1:33319        0.0.0.0:*                LISTEN      -
tcp        0      0 0.0.0.0:22             0.0.0.0:*                LISTEN      -
tcp        0      0 127.0.0.1:8998         0.0.0.0:*                LISTEN      -
tcp        0      0 0.0.0.0:971            0.0.0.0:*                LISTEN      -
tcp        0      0 127.0.0.1:40481        0.0.0.0:*                LISTEN      -
tcp6       0      0 :::22                  :::*                   LISTEN      -
tcp6       0      0 :::900                  :::*                   LISTEN      -
tcp6       0      0 :::922                  :::*                   LISTEN      -
tcp6       0      0 :::970                  :::*                   LISTEN      -
tcp6       0      0 :::981                  :::*                   LISTEN      -
tcp6       0      0 :::980                  :::*                   LISTEN      -

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yalinib884@cloudshell:~/linux_new/permission_practie$ ss -tulnp
Netid State  Recv-Q Send-Q Local Address:Port      Peer Address:Port      Process
tcp    LISTEN  0      511      127.0.0.1:46865        0.0.0.0:*                *
tcp    LISTEN  0      511      127.0.0.1:33319        0.0.0.0:*                *
tcp    LISTEN  0      128      0.0.0.0:22            0.0.0.0:*                *
tcp    LISTEN  0      5      127.0.0.1:8998         0.0.0.0:*                *
tcp    LISTEN  0      128      0.0.0.0:971           0.0.0.0:*                *
tcp    LISTEN  0      128      127.0.0.1:40481       0.0.0.0:*                *
tcp    LISTEN  0      128      [::]:22               [::]:*                   *
tcp    LISTEN  0      4096      *:900                  *:.*                      *
tcp    LISTEN  0      4096      *:922                  *:.*                      *
tcp    LISTEN  0      4096      *:970                  *:.*                      *
tcp    LISTEN  0      4096      *:981                  *:.*                      *
tcp    LISTEN  0      4096      *:980                  *:.*                      *

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processing triggers for man-db (2.12.0-1build1) ...
yalinib884@cloudshell:~/linux_new/permission_practie$ traceroute google.com
traceroute to google.com (74.125.200.113), 30 hops max, 60 byte packets
 1  10.88.0.1 (10.88.0.1)  0.069 ms  0.021 ms  0.013 ms
 2  * * *
 3  172.253.67.216 (172.253.67.216)  0.528 ms * *
 4  * * *
 5  * * *
 6  * * *
 7  * * *
 8  * * *
 9  * * *
10  * * *
11  * * *
12  * * *
13  sa-in-f113.1e100.net (74.125.200.113)  0.586 ms * *
yalinib884@cloudshell:~/linux_new/permission_practie$ wget https://example.com/file.zip
--2026-05-14 17:58:37--  https://example.com/file.zip
Resolving example.com (example.com)... 104.20.23.154, 172.66.147.243, 2606:4700:10::ac42:93f3, ...
Connecting to example.com (example.com)|104.20.23.154|:443... connected.
HTTP request sent, awaiting response... 404 Not Found
2026-05-14 17:58:37 ERROR 404: Not Found.

yalinib884@cloudshell:~/linux_new/permission_practie$ curl https://google.com
<HTML><HEAD><meta http-equiv="content-type" content="text/html; charset=utf-8">
<TITLE>301 Moved</TITLE></HEAD><BODY>
<H1>301 Moved</H1>
The document has moved
<A HREF="https://www.google.com/">here</A>.

```

```

yalinib884@cloudshell:~/linux_new/permission_practie$ nslookup google.com
Server:      169.254.169.254
Address:     169.254.169.254#53

Non-authoritative answer:
Name:   google.com
Address: 74.125.200.139
Name:   google.com
Address: 74.125.200.102
Name:   google.com
Address: 74.125.200.101
Name:   google.com
Address: 74.125.200.100
Name:   google.com
Address: 74.125.200.138
Name:   google.com
Address: 74.125.200.113
Name:   google.com
Address: 2404:6800:4003:c0f::66
Name:   google.com
Address: 2404:6800:4003:c0f::8b
Name:   google.com
Address: 2404:6800:4003:c0f::8a
Name:   google.com
Address: 2404:6800:4003:c0f::65

```

```
yalinib884@cloudshell:~/linux_new/permission_practie$ dig google.com

; <<>> DiG 9.18.39-0ubuntu0.24.04.3-Ubuntu <<>> google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 6758
;; flags: qr rd ra; QUERY: 1, ANSWER: 6, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;google.com.                IN      A

;; ANSWER SECTION:
google.com.                291     IN      A      74.125.200.139
google.com.                291     IN      A      74.125.200.102
google.com.                291     IN      A      74.125.200.101
google.com.                291     IN      A      74.125.200.100
google.com.                291     IN      A      74.125.200.138
google.com.                291     IN      A      74.125.200.113

;; Query time: 1 msec
;; SERVER: 169.254.169.254#53(169.254.169.254) (UDP)
;; WHEN: Thu May 14 17:59:04 UTC 2026
;; MSG SIZE rcvd: 135
```